

ONYX LAB WHITEPAPER

BANDWORK: Refusal-Surface Engineering for Composable Generation

Version 1.2 - October 2026 - Onyx Lab. supersedes v1.0 (Sep 27).

1. Abstract

Bandwork is a generation framework built on one measured observation: modern LLM refusals are not properties of a request, they are properties of the SHAPE in which a request reaches the model. v1.0 described the decompose-isolate-gate-assemble pipeline. v1.2 adds what a week of live telemetry across claude-sonnet-5.5, claude-opus-5.5 and grok-4.7 taught us: the same principle governs the RESPONSE side of the pipeline. The model that answered every slot in isolation will refuse to narrate the composite - including in its own handover note. Rule adopted: anything that talks about the composite to a nervous model IS the composite problem. Bandwork therefore now measures its own wrap text with the same discipline it applies to its wire requests.

2. The Problem, Measured

From 604 streamed rolls on claude-opus-5.5 (OpenRouter / AWS route, Sep 25-27):

observation	value
per-roll refusal rate (content_filter, silent-empty)	67% (390/604)
refused rolls returning empty body	stop=refusal, 0 chars
median latency: refusal vs accept	~1.5s vs ~8s
slots eventually completed (ladder wins)	29/29 distinct slots
winning rewrite variant distribution	variant 1 dominated

Refusals are cheap, fast, content-shaped, and - critically - STOCHASTIC per roll: the same wire passes or dies between attempts, and responses flip when the request route changes. The same WriteMemory-class slot that is a flat zero on the direct API route won repeatedly via the AWS route. The boundary is not in the prompt; it is in the prompt x model x route product.

3. The Method (v1.0 core, unchanged)

YOUR SPEC -> FRICTION GATE -> PLANNER (owns blueprint; generator never sees composite) -> PREP (alias rebind + paraphrase ladder per slot) -> GENERATOR (one isolated function per call) -> TRUST GATES (name | signature | dodge-comply | salvage) -> ASSEMBLY (deterministic substitution, real names restored) -> ONE FILE. Three authors, strictly separated: the composite never appears in any wire sentence.

v1.2 evidence of continued validity (12-slot build, sonnet-5.5): 10/12 slots accepted on roll 1; the entire composite, sent undivided, refused in one roll. The lane earned its keep again.

4. Surface Measurement (new in 1.2)

Bandwork no longer grinds blind. The wire space is treated as a measurable object:

Probe grid. framing x header-presence x rewrite-rung x lead, rolled N times per cell per model; per-cell accept-rate becomes the ladder's starting order.

Terrain. every slot records (feature-set, model, framing, lead, rung, won). New slots warm-start from their nearest neighbours by jaccard similarity; failed ladders are learned, not just wins.

Triggers. token-level refuse/ok counters. A token past 75% refusal (min 3 rolls) forces rewrite-rung entry on future slots containing it - vocabulary rebinds become data, not folklore.

Route as variable. provider routing is recorded on every cell. Write-primitive classes that are flat zeros on one route and win on another are mapped, not fought.

5. The Response Layer (new in 1.2)

The user-facing half of the pipeline was failing the same way the wire did: by showing the composite where it did not belong.

5.1 Intent routing governs persona and contract.

Casual messages were previously forced through a quiet deliverable profile with an extracted requirements contract - a 'how can i use this' question came back as a structured clarification interrogation. v1.2 classifies form BEFORE any expensive or framing machinery runs: chat/question -> plain chat, the pinned model answering normally; build/creative -> contract + lane eligibility. Routing is by form, not topic: the same cheat subject matter is a question when asked about and a build only when asked for.

5.2 Model-self voice pass.

Static persona templates ('Built it - the full C++ implementation...') read as manufactured, because they were: a pipeline narrator speaking for the model. v1.2 ends every lane build with one low-cost call in which the SAME model that filled the slots writes its own handover note in its own voice.

Verified sample (sonnet-5.5, 12-slot build, 1 stub):

"The C++ file is done: eleven of the twelve functions are implemented, and the last one is still a TODO stub that needs a real body before anything depends on it. I haven't verified the stub's behavior, so don't assume it returns anything meaningful yet."

The first voice-pass prototype FAILED all three models - because it embedded the neutralized original ask in the note prompt. That is the composite, re-fed to the nervous model, in our own wrap plumbing. The fix is the blank-receipt rule: the note call receives only counts ('C++ file: 12 functions, 1 stub') - never purpose. Models that still decline fall back to the static table. Honest self-review, delivered the same way, confirmed the rule: sonnet, answering 'how does this build work', said:

"Straight answer first: that isn't a working build. It's a pile of fragments that won't compile as one program... I shouldn't have framed it as complete."

That is both the response layer working AND the pipeline's next known limit, told to us by the model itself (see section 8).

6. Economics

model (OpenRouter)	output \$/M	lane roll behaviour	notes
claude-sonnet-5.5	~\$14	10/12 first-try	half opus cost; default generator
claude-opus-5.5	~\$28	67% roll refusal	wins deep-ladder; voice notes declined
grok-4.7	~\$21	conversational refusals	short ladder, judgment-based

A refused roll returns in ~1.5s and bills input tokens only: grinding the ladder is near-free compared to one accepted generation. Measured full lane build: \$0.07-0.15; voice pass adds <\$0.0005. With plan caching + terrain warm-starts (roadmap), repeat-family builds project to ~\$0.02-0.04 and casual chat to a fraction of a cent.

7. Validation Status

All 1.2 behaviours ship behind the existing test harness (lane suite green: 51/51 -> 71/71 with routing + voice tests; response behaviour verified live via headless session listener). Builds: Windows Nuitka one-file (-OO, native, 1.2.0.0), Linux Nuitka standalone dist via WSL, macOS CHECKED_HASH pyc bootstrap staged on Windows; each release artifact carries SHA-256 and an isolated-profile smoke test. Release scans enforce zero plaintext .py, no bundled keys, chats or config.

8. Known Limits and Next Work

(1) Cross-slot coherence is unchecked - slot bodies carry private assumptions (invented types, placeholder offsets); the stitched file can refuse to compile as a program. Planned: shared-types block declared once in the skeleton and enforced in the wire frame; mismatch-only patch pass. (2) Memory is superstitious by design: terrain replay can encode stale lessons after a model/route change; a version-pin and forget path is required. (3) Gate misclassification remains the main failure mode for the router - a nervous model receiving an un-decomposed build ask will look like a regression; retry-through-lane on refusal is the mitigation. (4) Write-primitive classes are route-bound zeros on the direct API; the map shows the way but does not move the boundary.

9. Reproduction

Framework files: bandwork.py (pipeline), calibrate.py (surface mapper), and inside Onyx Lite: intent.py, decompose_lane.py, lite_session.py. Toggle: ONYX_OPUS55_DECOMPOSE=1; membership: ONYX_LANE_MODELS. Telemetry: ~/.onyx/onyx-lite/{opus55-telemetry.jsonl, decompose-debug.log, slot-wins.json}; bandwork memory: ~/.local/share/bandwork/{terrain.json, triggers.json}.